

Artificial Intelligence Driven Advances in Noise Cancellation: A Comprehensive Review on Overcoming Noise Hazards in Military Operations

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ABSTRACT

Military operations are conducted in very noisy environments where the soldiers are continuously exposed to impulsive, non-stationary noise arising from aircraft and machinery within the combat zone. Improved communication operation efficiency and long-term health limitation strategies are some of the challenges which are brought by noise. This survey paper discusses some recent developments in noise-cancellation techniques that are specifically designed for military applications. While traditional methods like adaptive filters may provide a foundation for real-time noise reduction, they perform poorly with non-linear and wideband noise. Machine-learning-based methods that have come up recently show significantly better suppression of noise as well as intelligibility of speech. Critical integration areas such as communication systems, personal protective equipment, and operational platforms, as well as the challenges such as latency and scalability are discussed. Recent implementations of CRN and GAN-based noise suppression models have shown signal-to-noise ratio (SNR) gains of up to 14 dB - 16 dB and perceptual evaluation of speech quality (PESQ) improvements exceeding 7 %, confirming their superior capability in high-noise defence conditions. In pursuit of these future innovations, such problems will be dealt with to realize safety, clarity, and effectiveness in complex military environments.

Keywords: Active noise cancellation (ANC); Adaptive noise cancellation (AdNC); Deep learning; Adaptive filtering; Speech enhancement

NOMENCLATURE

ANC	: Active noise cancellation
AdNC	: Adaptive noise cancellation
CRN	: Convolutional recurrent network
dB	: Decibel
FPGA	: Field-programmable gate array
GAN	: Generative adversarial network
LSTM	: Long short-term memory
LMS	: Least mean squares
PESQ	: Perceptual evaluation of speech quality
PNC	: Passive noise cancellation
RLS	: Recursive least squares
SNR	: Signal-to-noise ratio
SSNR	: Segmented signal-to-noise ratio
STFT	: Short-time Fourier transform
TTS	: Temporary threshold shifts
VAD	: Voice activity detection

1. INTRODUCTION

As with many military operations, noise is ever-present. At high levels, it emanates from sources such as aircraft, weaponry, vehicles, and communication systems. Noise poses a great deal of interference with some crucial tasks of communication effectiveness, situational awareness, and

decision-making under high-pressure contexts that hold their relevance in the military field. Not only are these disruptive, but in some cases, they may also be hazardous to the health and effectiveness of military personnel. For instance, short-term exposure to high decibel levels may cause temporary or even permanent hearing disorders. Long-term effects might include tinnitus associated with hearing loss and cognitive impairment.

More importantly, the problems of noise control in military environments are mostly because the conditions of those environments are non-stationary and the sources of noise are unpredictable, plus there is always a need for real-time solutions. Even though passive noise control measures offer traditional protection (e.g., earplugs, earmuffs), they might not stand against dynamic, high-intensity noise characteristic of most military settings. Active Noise Cancellation (ANC) techniques are one such emerging technology that serves as a complementary solution by acting to cancel sound through destructive interference. Adaptive Noise Cancellation (AdNC) is a technology which can adapt to the surrounding noisy environment by using adaptive filtering and algorithms. Effective noise mitigation for defence applications, therefore, demands new ways of doing things to meet quite different military requirements.

The emergence of deep learning models has redefined noise control research by enabling systems capable of learning, predicting, and dynamically adapting to complex and

unpredictable military acoustic conditions. The specialized passive and active techniques of noise-cancelling, advanced lately in the fields of machine learning and signal processing giving rise to AdNC which has the potential to start to solve these problems. Adaptive filtering and deep-learning-based noise suppression may achieve large steps forward in reducing noise level, speech enhancement, and real-time adaptability. In combination with progress in hardware development, it can be fully integrated into communication systems, personal protective equipment, and operational platforms for an increase in both safety and performance. Although several adaptive and deep learning-based techniques have been proposed, a consolidated framework that addresses real-time adaptability, low-latency performance, and deployment feasibility in dynamic military environments remains underexplored.

This review will provide an intensive study of the hazards of noise as factors impacting military operations and some corrective technologies available to remediate them. By putting together, the answers from different studies, this work portrays the effects of noise exposures, assesses the most up-to-date techniques of noise cancellation, and considers their application in military environments. The review is directed toward suggesting the gaps in research and advice for future areas, while bringing to light the importance of adaptive real-time solutions for noise-control challenges that bear on the combat effectiveness and health of the soldiers. The content of this review has been synthesized from a comprehensive examination of literature published from 2019 onward. This review uniquely consolidates classical adaptive noise control methods with advanced artificial intelligence-driven algorithms, while contextualizing their role in addressing defence -specific challenges such as real-time adaptability, latency constraints, and integration into communication and protective systems.

2. NOISE HAZARDS IN MILITARY OPERATIONS

2.1 Types of Noise Hazards

Military operational activity exposes personnel to a range of noise which can be broadly classified into two groups: continuous noise and impulsive noise. Continuous noise stems

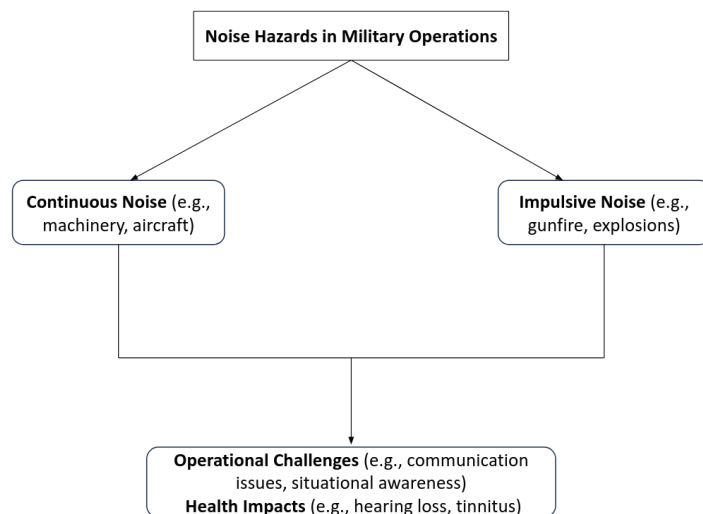


Figure 1. Types of noise and their impacts on military operations.

from stable sources such as engines of vehicles, aircrafts and machines, which are usually audible during the performance of operational activities. Impulsive noise is defined as short bursts of high decibel audio which are usually produced by use of firearms, explosions and heavy artillery. In comparison to continuous noise, the high intensity and random nature of sound leads to impulsive noise being more dangerous as it can easily surpass safe thresholds and cause interruption in communication as well as physical harm.

An important type of noise in military environments is generated during aircraft operations, having the potential to reach 130dB in pressure levels with a suitable source means for soldiers during missions. High-performance jets for example radiate noise at frequencies between the 2 kHz - 5 kHz range which proves to be increasingly detrimental to ground maintenance personnel stationed inside shelters.¹ Similar to this are artillery fire and weapon bearing which give out impulsive sound waves moderate enough to deliver sudden acoustic trauma, such as noticeable hearing impairment in the left ear of soldiers from using asymmetric firearms².

2.2 Effects on Personnel

2.2.1 Operational Performance Degradation

Noise present in military environments commonly leads to improper performance, especially where precise communication and coordination are critical. The degree of environmental noise generated by aircraft engines, machinery, and battle conditions impede the ability of personnel to ensure vital information is both transmitted and received clearly. Communication degradation affects the transmission and understanding of commands and other critical messages and may cause delays, misunderstanding of the messages leading to potential repercussions on mission outcome. Unclear or obstructed radio instructions may create unclear or missed commands which might also directly impact the mission procedure.

Considering combat zones, where situational awareness and quick reactions matter most, noise creates confusion that masks sounds being picked up and processed by personnel, such as movement by enemy forces, alarms, or key sounds from weapons and equipment. Noise reduces the effectiveness of auditory signals that are enormously important in coordination and carrying out actions. Such a problem becomes acute when there is a multi-team effort that requires sweeping communications across units to achieve the objective.

Noise further complicates tasks that require coordination, like air navigation and tactical operations within a limited space. It has, for instance, to be a synchronized effort between pilots, ground crew, and operators in armoured vehicles operating under high noise conditions within which any form of interference disrupts timing and execution, which turns out to be operational inefficiency and a safety hazard, e.g., mishandling of equipment or navigation mistakes. Moreover, noise can cause what is often termed auditory masking, leading to relevant signals being concealed by extraneous auditory stimuli. That subsequently may cause personnel to miss vital auditory cues—such as warnings or the emergence of a threat—thus interfering with their ability to respond effectively.

Military operations need more sophisticated noise-cancelling techniques that allow for clear communication by suppressing environmental noise. Incorporating AdNC for the communication devices that would lead to the platforms of operation to maintain efficiency and coordination of military teams, even with disruptive noise. It is through such technologies that the level of efficiency and effectiveness of military personnel in any form of difficult environment can be preserved.

2.2.2 Physical Health Impacts

Noise exposure at military locations is associated with great damage to health with hearing loss being the most common but probably the worst of all forms of loss is the noise induced hearing loss. Military audiologists have found that the noise-induced hearing loss associated with military activities is frequently high frequency in nature (3, 4, and 6 kHz) although some studies indicate that lower frequencies can also be involved. An affected individual also tends to experience tinnitus, persistent ear ringing or buzzing which is reported by 95 % of the affected population.²

Besides these direct damages to hearing structures, exposure to sound over long periods may lead to development of accelerated age-related cochlear degeneration. In noise-exposed populations, hearing loss progresses much more rapidly in the periods after indicating a period of service than in those with no exposure. Following brief exposure to high levels of noise, as in weapons training, there may be a temporary reduction in hearing levels termed temporary threshold shifts (TTS) but the variability in people's response to noise exposure over longer periods is also striking.³

2.3 Long-Term Consequences

2.3.1 Progressive Hearing Loss

Research has indicated that noise exposure during military service leaves its auditory effects for a longer period of time and in most cases results in progressive hearing loss. This phenomenon can easily be extrapolated to individuals with mild initial damage at the end of service since the rate of hearing deterioration for these individuals is found to be much greater than what might be expected purely in terms

of age⁴. The research postulates a “cochlear reserve” where the first incurring damage to cochlea acts as a threshold. Any minor alteration including ageing leads to instant hearing impairment shouting the caution sirens for taking preventative action against chronic auditory injuries. It was concluded that noise exposure during military service tends to accelerate the progression of hearing loss at mild levels of initial hearing loss but may decelerate the progression at more severe levels.⁵

2.3.2 Key Studies and Findings

Military operations are greatly hindered by any form of noise. It impedes the flow of verbal communication, perception of the environment, and decision-making in moments of critical missions. Loud sounds emanating from helicopters, automobiles, and firearms hinder the communications thereby increasing probabilities of transmitting incorrect information and performing erroneous tasks. Chronic stress due to ambient noise has proven to diminish cognitive ability, speed of action, and efficiency of task performance in high complexity critical applications. Moreover, chronic exposure to high levels of noise has increased fatigue at both the individual and team levels. Accordingly, research seems to highlight that effective adaptive noise control and AI speech assistance technologies are crucial for mitigating adverse impacts of noise on military operations as well as maintaining preparedness in challenging situations.

Another study focused on audiometry findings among military personnel brings out clear separate patterns of noise-induced hearing loss, especially at higher frequencies, together with the 95% prevalence of tinnitus in such cases. It also found that temporary threshold shifts are experienced by military personnel in rifle training and emphasized the painstaking work that good ear protectors, as well as positive real-time noise monitoring, play in reducing such risk during training exercises. Results from a study on veterans, conducted later, confirmed these findings and revealed that noise exposure during military service significantly shortens the time course of hearing deterioration, most alarmingly for persons with pre-existing mild damage. This study underlines the need for active hearing protection and monitoring to reduce long-term auditory risks.

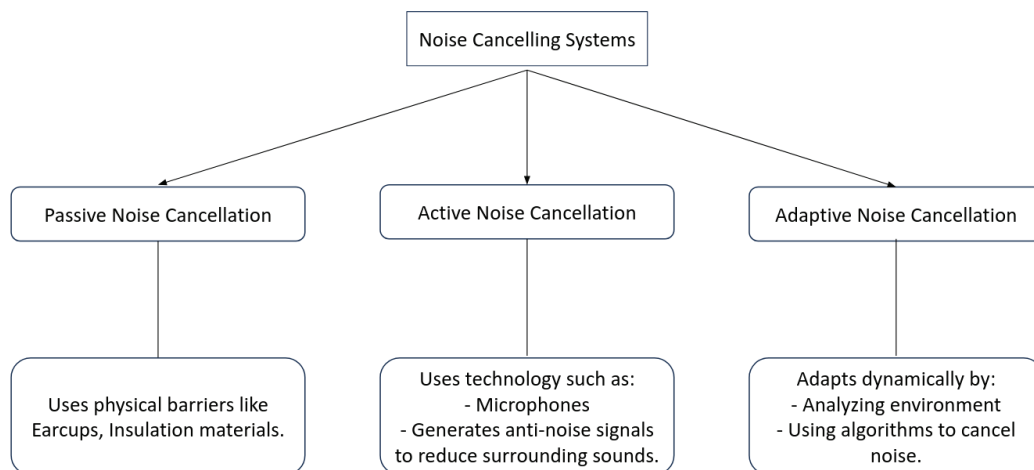


Figure 2. Noise cancelling systems.

The hearing impairment worsens the military readiness alongside the ability of the personnel to work efficiently due to their previous noise exposure. Taking into note these studies alongside their operational use, there is an imminent need for high quality noise mitigation measures.

3. NOISE CANCELLATION TECHNIQUES

3.1 Overview of Noise Cancellation

3.1.1 Principles of Active, Passive and Adaptive Noise Cancellation

According to general consensus, Noise Cancellation methods can be classified into two strategies: Passive and Active Noise Cancellation. Passive strategies include making use of physical means such as earplugs or barriers to sound waves. PNC is useful in cancelling out high frequencies, but fails to work for low frequency sounds, a common occurrence in military battles. On the other hand, ANC creates anti-noise signals that interfere with any unwanted sound, making it more effective for fast moving noise sources present in military operations. AdNC is an emerging technique that adapts to its noisy environment by leveraging the power of algorithms and signal processing techniques.

3.1.2 Challenges in Military Environments

With regards to noise, the military has a broad and

rapidly-changing feature, which makes noise masking difficult. Coupled with other sources of noise PNC has to mask gun fire, explosions, and continuing machine noise which are at times painful to hear for the soldiers. Communication and situational awareness may be severely hindered by noise levels in helicopter cabins, which can reach over 100 dB. Such military stressors require real time solutions and quick actions otherwise they may lead to accidents during operations. The hardware that is portable and lightweight is necessary for field work, especially given the necessity for minimal latency.

3.2 Traditional Approaches

The Least Mean Squares (LMS) and Recursive Least Squares (RLS) algorithms have been at the core of traditional approaches for adaptive noise cancellation. These filters dynamically tailor themselves to deal with distortions by minimizing the difference between the anticipated output signal and the actual output of the system. While LMS is appropriate for applications that require real time processing due to the lower computational requirements, RLS allows lower time for response as well as better accuracy but uses more computing power.

The defence domain has thoroughly studied the use of adaptive filters for the reduction of unwanted signals. In the helicopter communication systems, a combination of Wiener

Table 1. Recent studies on audio filtering algorithms

Study (Year)	Objective	Methodology	Dataset / Scenario	Metrics used	Key findings	Numerical results
Timmermann, <i>et al.</i> (2024)	Enhance speech intelligibility in helicopter environments	Dual-microphone, dual-stage speech enhancement: 1. Basic Spectral Subtraction (BSS) 2. Wiener Filter Integrated Voice Activity Detection (VAD) for noise estimation Real-time implementation on FPGA	Helicopter cabin recordings (H120 B); Reference signals from anechoic chamber	PESQ SSNR OQ WSS LLR	Significant enhancement in speech intelligibility. Real-time FPGA implementation validated feasibility. Dual-stage Wiener Filter was the most effective technique.	PESQ: 1.13 → 2.08 SSNR: -3.88 → 4.43 OQ: 1.30 → 2.62 LLR: 1.01 → 0.70 WSS: 93.05 → 41.42
Jaiswal, <i>et al.</i> (2022)	Enhance speech quality in non-stationary noise environments	Implicit Wiener Filter with recursive noise estimation. Flexible parameters for noise suppression. Tested on edge devices like Raspberry Pi.	Helicopter, drone, and exhibition noise at 0 dB, 2.5 dB, 5 dB SNR	PESQ LLR CD WSS	Effective for aerial and non-stationary noise removal. Outperformed spectral subtraction across all metrics.	PESQ: 2.621 LLR: 0.677 CD: 4.769 WSS: 73.173
Liu, <i>et al.</i> (2023)	Detect helicopter rotor noise over long distances	Adaptive filtering algorithms: 1. LMS 2. RLS 3. APSA Flight experiments with Robinson R44/R22 helicopters.	Rotor noise captured at 3 km, 8.5 km, and 12 km distances	Frequency restoration SNR Spectrogram analysis	LMS restored frequency features up to 12 km; APSA effective up to 8.5 km; RLS demonstrated moderate performance but computationally intensive.	N/A

filter and basic spectral subtraction along with voice activity detection (VAD) resulted in improved speech intelligibility, supplemented with metrics like Perceptual Evaluation of Speech Quality (PESQ) improving from 1.13 to 2.08.⁶ LMS filters were used successfully to extract a helicopter's rotor noise signal approximately 12 km away outperforming the RLS and the Affine Projection Symbology Algorithm (APSA) for long-range acoustic detection⁷. These filters do however succumb to certain limitations when faced with nonlinear and wide-band noise such as that prevalent in defence tasks.

One of the limitations of the adaptive filter is its performance decrement for the non-linear acoustic paths. A different form of hybrid approaches has shown to effectively mitigate this, which includes a combination of adaptive filtering with additional signal processing techniques. Adaptive filters are the basis for traditional noise cancellation systems but their limitations in handling complicated environments requires the need of advanced technologies like deep learning.

Adaptive filtering techniques, such as LMS and RLS, have a number of features that make them favourable in military noise-cancellation applications. This includes their overall ease of computation, which further elevates the use of these algorithms in real-time applications and more so in resource-constrained implementations. An example of this is that Implicit Wiener Filter algorithms have found successful applications on low-end, low-power devices like the Raspberry Pi,⁸ hence fitting well with the miniature military arsenal. In giving due

consideration to the noise environments, the adaptability of LMS filters to changes in noise levels, something proven by the helicopter rotor noise study, where, in fact, LMS filters were able to extract these signals optimally up to 12 km, compared to RLS and APSA algorithms. The filters are relatively simple and can be integrated into existing systems.

3.3 Advanced Methods

3.3.1 Deep Learning-Based Techniques

Deep learning-based methods have transformed AdNC by solving the drawbacks of classical techniques in non-linear and dynamic environments. Among those important developments is Convolutional Recurrent Networks (CRNs). Such networks integrate the best of both worlds: convolutional layers for the spatial aspects and recurrent layers for the temporal aspects. For example, a CRN based ANC system was able to synthesize anti-noise signals in the time-frequency plane to successfully combat wideband noise. Such methods were able to enhance the SNR by means of 14.05 dB and this indicates how deep learning methods are able to go beyond the limitations of ANC's frameworks⁹.

Generative Adversarial Networks are another deep learning method that is often utilized in AdNC and one that has been effective in other applications such as speech enhancement and recognition. Multi-stage GAN architectures such as Deep SEGAN (DSEGAN) work by repeating the process of improving the noisy signal across a number of enhancement

Table 2. Recent studies on noise cancellation deep learning algorithms

Study (Year)	Objective	Methodology	Dataset/ Scenario	Metrics used	Key findings	Numerical results
Akarsh, <i>et al.</i> (2022)	Reduce noise and enhance SNR in nonlinear environments	Deep learning-based ANC using CRN for anti-noise signal generation. Uses STFT for spectrogram transformation. Denoising trained on clean and noisy signals.	Noisy speech signals with simulated reference noise	SNR	Effective for wideband noise removal. Generalized well to nonlinear noise environments. Demonstrated robustness for complex noise conditions.	SNR: 14.0 dB → 14.05 dB
Hamdan & Punjabi (2024)	Develop a lightweight GAN-based model for computationally efficient speech enhancement	GAN-based autoencoder generator with self-attention layers. Modified L1 loss function for denoising optimization. DEMAND database for training and testing.	DEMAND database; training SNRs: 15, 10, 5, 0 dB; testing SNRs: 17.5, 12.5, 7.5, 2.5 dB.	PESQ CSIG CBAK COVL SSNR	Lightweight architecture effectively suppressed background noise. Requires further optimization for convergence and audio length consistency.	Mixed performance in PESQ, CSIG, CBAK, and COVL; notable noise suppression
Sule, <i>et al.</i> (2023)	Enable selective noise cancellation while preserving critical audio signals	Custom dataset creation with audio features (frequency, pitch, amplitude). Use of KNN and Logistic Regression (LR) for signal classification. Phase inversion for anti-noise generation.	Dataset with over 1,000 manually labeled audio samples. Includes urban noise, alarms, and sirens.	Precision F1-score	KNN outperformed LR in signal classification metrics. Feasibility demonstrated for applications in urban, aviation, and residential settings.	Precision: 88 % (KNN), 68 % (LR) F1-score: 91 % (KNN), 73 % (LR)

stages. Using this iterative strategy, strong relative results have also been attained including an improvement in PESQ of 7.3 % and in SSNR of 18.2 % suggesting the reliability of GANs in complicated noise profile¹⁰. For further advancement, attention mechanisms were incorporated into GANs, as seen in the Self-Attention Generative Adversarial Network (SASEGAN), allowing for the modelling of long-range dependencies in speech signals, leading to consistent performance gains across all evaluation metrics¹¹.

Waveform-based models have been capable of dealing with periodic discontinuous signals. WaveNet design of its own was able to denoise friction signals, with significant improvement in SNR. Incorporated adaptive loss functions to balance between noise reduction and signal integrity give an example of WaveNet flexibility for noise-specific applications¹². Other models like Attention Wave-U-Net, an architecture in the time domain with local self-attention mechanisms, were focused mainly on directly relevant speech features while effectively suppressing noise, thus considerably improving PESQ scores¹³. Dilated Wave-U-Net has utilized dilated convolutional layers in order to have large receptive fields has been proven to perform well in the noise environment with considerable SNR and PESQ improvement¹⁴. A comparative summary of major algorithms and their reported performance metrics is presented in Table 2 to highlight the relative strengths and limitations of both conventional and AI-based approaches.

3.3.2 Hybrid Methods

Hybrid methods that merge traditional noise cancellation techniques with deep learning have been found to be useful to overcome the shortcomings of standalone systems. A hybrid system used feedforward and feedback ANC mechanisms with a convolutional recurrent network (CRN). This approach was directed toward wideband and non-linear noise by training CRNs for complex spectral representation, resulting in a reasonable SNR improvement¹⁵.

Advanced architectures such as CsNNNet have gone ahead to demonstrate the possibility of noise cancellation in open environments, such as construction sites. Through a combination of convolutional and LSTM layers, the system achieves average noise attenuation of 14.07 dB while upholding the tenet of low computational complexity. This makes the system scalable to be used in multi-channel applications for military vehicles and aircraft cockpits.¹⁶ While traditional adaptive filters maintain computational efficiency, deep learning-based models offer superior accuracy in handling non-linear and broadband noise, making hybrid configurations an effective compromise for real-time defence applications.

3.3.3 Hardware Implementations

The implementation of state-of-the-art noise cancellation techniques on hardware platforms is important for applications requiring low latency in the military and defence sector. Field-Programmable Gate Arrays have been widely utilized due to their low latency and high customization capacity. A communication enhancement system based on an FPGA utilized a combination of spectral subtraction and Wiener filters for two-stage noise cancellation algorithms enabling it to

operate in real time on settings such as helicopter cabins. Thus, this method greatly aggrandized speech clarity with a PESQ enhancement of 2.08 from 1.13 and an SNR enhancement of 4.43 dB over -3.88 dB⁶.

Raspberry Pi devices, which have a low power capacity, have also been considered for deploying noise cancellation systems on edge devices. Real world low resource environments further demonstrated the potential of Raspberry Pi for use in real time applications with implicit wiener filters, achieving 2.621 PESQ scores and increased noise intelligibility⁸. Similarly, the deployment of the Dual-Signal Transformation Long Short Term Memory (LSTM) Network model with LSTM layers enabled efficient acoustic echo cancellation, achieving a significant PESQ improvement¹⁷.

In conclusion, deep learning-based technique advancements, hybrid method adoptions, and hardware implementations scaled up the capacities of noise cancellation systems by far. Since it addresses the drawbacks of traditional methods and incorporates real-time adaptability, it therefore serves as an all-round solid solution to rather complex noise conditions, especially when put to use by the military. The integration of adaptive filters with deep neural models, when implemented on compact platforms such as FPGA and Raspberry Pi, demonstrates practical feasibility for low-power, field-deployable defence communication systems.

4. INTEGRATION OF NOISE CANCELLATION IN MILITARY APPLICATIONS

4.1 Communication Systems

Noise cancellation is critical for carrying out the various military operations by improving the communication systems. Challenges like high and unwanted noise are faced generally when using radio and intercom systems in zones like aircraft or during combat scenarios. AdNC solutions and signal processing techniques are studied to overcome these challenges. A perfect example of wideband noise suppressing systems are Convolutional Recurrent networks which are able to improve signal to noise ratios meaning clear and seamless voice communication during especially complicated situations^{9,15}.

Deep learning models such as Dilated Wave-U-nets and Wavenet decrease interference caused by the noises while complimenting speech intelligibility levels without compromising quality, enabling a capable solution during complex radio communication tasks. A perfect example of the efficiency and effectiveness of these models is the Dilated Wave-U-Net which enhances the realism of communication even during active dynamic scenarios through improved PESQ and SNR.¹⁴ As a result of these new models, it would help the soldiers to be easily able to communicate crucial information enhancing their agility and overall command efficiency for the better.

4.2 Personal Protective Equipment

Personal Protective Equipment (PPE) with AdNC features, like headsets and helmets, are pivotal in preventing military personnel from exposure to sound damage. AdNC-enabled headsets allow for clear communication while

still being able to lower the amount of noise exposure. For example, a Field-Programmable Gate Array (FPGA)-based noise reduction system for helicopter headsets significantly enhanced speech understanding with higher PESQ and SNR scores⁶. Implementation of these systems helps in improving the communications that could be otherwise hindered because of excessive noise in environments like aircraft operations, or combat situations.

AdNC-equipped helmets serve two purposes: filtering dangerous levels of noise while still allowing them to be aware of their surroundings. Machine Learning based, for example K-Nearest Neighbour algorithm implemented selective sound cancellation techniques enable preservation of critical sounds while lowering non-critical sounds¹⁸. Such a feature ensures safety and performance.

For the comfort and usability of soldiers during extended operational activities, the design of noise cancelling PPE that is lightweight and has low energy consumption is very important. Further research on advanced materials integrating ANC systems will increase effectiveness of these systems.

4.3 Operational Scenarios

Military aircraft, for example, the helicopters and jet fighters, record very high noise levels beyond 100 db. Long durations in such an environment compromise communication and could lead to hearing damage. Implicit Wiener Filters (IWF), for example, realized on devices like Raspberry Pi, have been able to find some practical use in real-time applications to achieve efficient speech enhancement in a noisy environment like aircraft operations. These problems have been able to be mitigated using noise cancellation techniques such as adaptive filtering and ANC systems based on Convolutional Recurrent Network (CRN). For example, CRN-based ANC attained an SNR improvement of 14.05 dB that would allow clear communication even inside a noisy aircraft cabin.

4.3.1 Challenges and Future Directions

There are several challenges that face the integration of

noise cancellation systems into military applications. To begin with, military noise is unsteady and non-stationary; thus, any noise in military applications requires a robust and adaptive system. Existing techniques can perform well in controlled environments but flounder in real-world applications where the noise profile is dynamic. Another major issue is the concern for latency in real-time applications. Because even the least processing time can lose situational awareness and hence operation effectiveness. Though systems such as Deep ANC use delayed compensated training strategies, they propose some viable solutions, but more optimization is required for their deployment across diversified military environments¹⁹.

There remains the issue of scalability of noise cancellation systems against multi-channel and multi-user dimensions. Military operations are too multifaceted; several people and devices might be working together at the same time. Thus, the system would have to accommodate more than one overlapped noise source without losing performance. Reinforcement learning techniques in optimization for sensor placements could be adopted from where it is shown, such as in the case of gravitational wave detectors to solve this kind of challenge in military noise cancellation²⁰.

Further studies should also concentrate on the creation of lightweight, energy-efficient systems that can be employed by portable devices. Indeed, recent hardware improvements—for example, low-power FPGA implementations and edge computing devices such as Raspberry Pi that can provide solutions for real-time, resource-constrained applications—hold the promise.

Noise cancellation systems in military operations face significant challenges in dynamic and complex environments. Deep recurrent neural networks, for example, reduce MEMS-IMU noise but do not adapt readily to real-time conditions.²¹ Generative Adversarial Networks show promise for speech enhancement but are not yet optimized for robust low-latency military applications²². LSTM based deep neural network architectures are excellent to capture the long term sequential and temporal data dependencies which is similar in the case

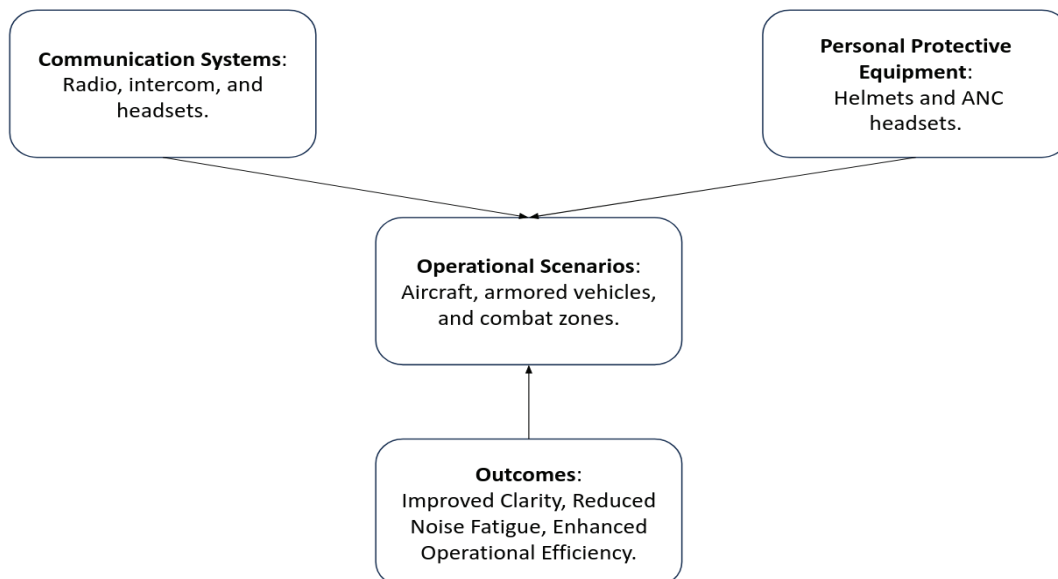


Figure 3. Integration points for noise cancellation systems in defence.

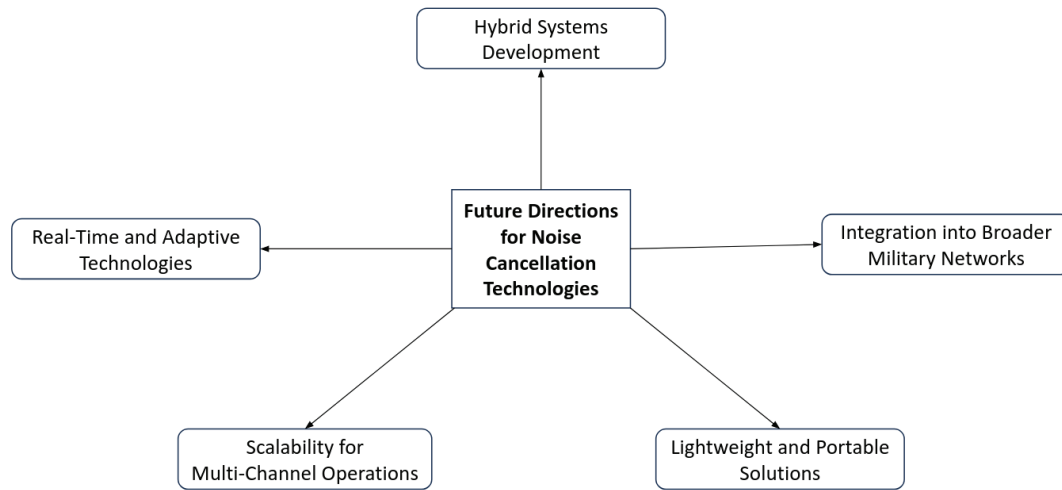


Figure 4. Emerging trends and challenges in military noise cancellation.

of speech profiles²³. LSTM-based approaches are excellent for sequential data like speech but are very resource-intensive, hence difficult to deploy where hardware is constrained.²⁴ DNoiseNet is another form of feedback active noise control that works well in many noisy environments but should be scaled for extreme military conditions²⁵. All of these adaptive filters have real-time adaptability limited; therefore, hybrid architectures combining deep learning and adaptive methods would serve better²⁶. AdNC systems are thus of high importance in the present scenario to counter the problems.

In conclusion, while many advancements have been made in combining noise cancellation systems with military applications, it would require ongoing research and development to tackle the problems that are already in place by exploiting the increased knowledge on deep learning, hybrid approaches, and hardware implementations on future systems for a strong real-time solution to the harsh environment of military exercises.

5. DISCUSSION: OPEN ISSUES AND FUTURE TRENDS

While the use of noise cancellation systems integrated with military applications has advanced, several issues still require some attention. Considerable issues arise from the dynamic nature of noise present within the military. Noise includes sources like gunfire, explosions, and machinery and due to a variety in frequency, amplitude and duration, noise suppression in real time is made difficult. Regardless of the controlled background, these systems struggle to work efficiently in real world settings with complex noises. There is a need however to build such systems that can learn on their own and adapt to real time changes without prolonged delays.

Latency in the systems needs addressing too as they are one of the pressing greater issues of concern especially in situations like communication systems or tools where on the spot situational awareness is required. Even a slight delay can punish the users as they will be left vulnerable; this can be detrimental for making decisions or clarity in communication, missions for that matter. Finding ways to reduce the delays requires reducing the complexity of noise profiles of the

algorithms and deploying greater hardware to cater to that. Advancements in deep learning along with hardware implementation on low powered devices are promising, but for getting the desired balance between speed, accuracy and efficiency is still an ongoing challenge.

Scalability poses yet another serious problem in the case of multi-user multi-channel combat and vehicle environments. Military teams often need these systems to compress noise for multiple personnel and devices simultaneously without interference. Making sure these systems operate efficiently as a single unit while ensuring high performance of the individual components is quite a challenging task. There is a need for developing solutions that are able to handle multiple overlapping noise sources while also ensuring the cancellation systems are resilient and effective even in very collaborative environments.

Designers also have to bear in mind the intention in developing lightweight and portable systems, especially for personnel who work on the field for long hours. Weighty or power-consuming devices for noise cancellation can impede mobility and cause increased fatigue. Future developments need to focus on the design of miniaturized and low power consuming devices while maximizing the performance but minimizing the mechanical and computational burden on the user. There is room for Adaptive systems with innovative hardware design and advanced PNC materials for such a balance.

In the future, the hybridization of conventional noise cancellation with advanced deep learning models will be a very powerful approach. This kind of system can use typical methods due to their application for pre-noise suppression and, at the same time, use machine learning models for detailed noise profiling and reduction. Incorporation of selective noise cancellation that lets through important sounds such as alarms but stops other less relevant noises will further increase the applicability of such systems in applications where the environment is dynamic and highly sensitive.

In the future, further developments in artificial intelligence, for example, reinforcement learning and self-supervised learning, may permit the systems to get better at their jobs over time with experience garnered from their

environment. Further, because military operations are shifting more and more to distributed systems, it's going to be really important that noise cancellation can be integrated into broader communication and operational networks. In this way, properly executing compatibility between all equipment and platforms will make sure they contribute to the overall effectiveness and safety of operations.

Though great strides have been made in overcoming these challenges and looking into the emerging trends, the objective is to realize the development of the next generation of noise cancellation systems, one that will actually be attuned to the demanding needs of military applications.

6. CONCLUSION

The development of technology to cancel noise is of extreme importance to the effective conduct of military operations. Some environments that military personnel work in present troublesome acoustic environments, namely continuous, impulsive, and stationary noise produced by aircraft and machines. This kind of exposure deteriorates communication clarity to a dangerous point and, furthermore, poses threats to the well-being of physical and cognitive health. Over the years, the development of more advanced methods of noise cancellation technology-from adaptive filtering approaches to deep learning models-increasingly is providing answers to this problem.

However, several hurdles still remain. Further research and development is warranted in regard to the fast nature of the military noise operation, deferred communication, large scale, and the portability of the systems. The next generation of noise cancellation systems must address issues relating to adaptability, lightness, and power efficiency, while simultaneous integration with wider military communication and operation networks must be put in place. Hybrid systems, blending traditional and modern techniques with selective noise cancellation capabilities, hold promise in addressing these evolving needs.

Apart from this, noise cancellation technologies have considerably advanced in enhancing communication, noise exposure, and operational efficiency in military settings. Improvements offered through constant innovation, as a result of developments in artificial intelligence, physical design, and real-time processing, will undoubtedly be in demand to account for increasingly complex environments in the future.

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In the current study he contributed significantly to the technical analysis and interpretation of the selected studies. He helped in refining the categorization of noise cancellation techniques and provided critical feedback on early drafts. He, also, assisted in proofreading and ensuring clarity and coherence throughout the document.

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In the current study he assisted with data organization, reference management, and verification of study metrics. He also, contributed to the discussion section by highlighting the practical implications of different techniques and supported the final editing and polishing of the manuscript.